

Education

Duke University , Durham, NC Doctor of Philosophy, Chemistry Cumulative GPA 3.8/4.0	2015-2022
Texas Woman’s University , Denton, TX Bachelor of Science, <i>Summa Cum Laude</i> , Chemistry with ACS Certification Cumulative GPA 4.0/4.0	2010-2015
Chandler-Gilbert Community College , Chandler, AZ Cumulative GPA 4.0/4.0	2007-2010

Research Experience

Howard Hughes Medical Institute Janelia Research Campus, Ashburn VA

Advisor: Hari Shroff, Ph.D.

Overview:

- Developed microscopes from empty optical table
- Worked in collaboration with Misha Ahrens lab and Google on project 2.
- Worked in collaboration with theory fellow Magdalena Schneider on project 1 to implement deep neural network models.

Project 1: Phase Diversity Wavefront Sensing for Adaptive Optics in Fluorescence Microscope

Aims:

- Develop rapid, image-based adaptive optics technique which is broadly compatible across imaging modalities and facilitates high-resolution imaging at depth in living organisms.

Results:

- Built prototype widefield microscope and adapted algorithm developed in collaboration with Patrick La Riviere (U. Chicago) to test feasibility of phase diversity for fluorescence microscopy.
- Built asymmetric dual-view light-sheet microscope and extended method to measuring aberrations in 3D samples.
- Developed deep neural network model of deformable mirror behavior in collaboration with Magdalena Schneider, enabling acquisition of aberration-corrected images computationally, paving the way for hardware-free solutions for adaptive optics.

Importance:

- An easy, rapid method for estimating optical aberrations enables users to circumvent purchase and implementation of costly and difficult-to-use wavefront sensing devices.
- Broad compatibility across imaging methods enables broader accessibility of aberration-sensing and correction, enabling higher-quality imaging at depth.

Project 2: Two-Photon Light Sheet for Brain-Body Zebrafish Imaging

Aims:

- Develop whole-brain Zebrafish light sheet imaging system with two-photon excitation for improved image quality at depth and more biologically-relevant behavior results due to invisible NIR excitation.

Results:

- Designed and tested two-photon excitation modification to original Zebrascopes design

- Built and tested scope for imaging performance, phototoxicity.
- Led team of graduate students and technicians on build

Importance:

- This imaging system will enable whole-brain Zebrafish experiments where the sample is not perturbed by visible-light stimulus, which corrupts behavioral data results. This result will enable us to tie functional Zebrafish behavior data to connectome.

Duke University, Durham, NC

Advisor: Prof. Kevin D. Welsher, Ph.D.

Overview:

- Developed microscopes from empty optical table
- Spearheaded development for automation of all acquisition, processing, and visualization.
- Mentored 2 undergraduate students.
- Led a 4-person team on the 3D-TrIm project consisting of myself, a postdoctoral scholar, junior graduate student, and undergraduate student.

Project 1: 3D-FASTR – Rapid Point-Scan Volumetric Microscopy

Aims:

- Develop new microscope capable of rapid volumetric Imaging a moving sample

Results:

- Conceived theory of volumetric point-scan using tessellating pattern
- Developed 3D-FASTR volumetric two-photon laser-scanning microscope
- Performs 3D point-scan imaging 4-8x faster vs. conventional z-stack

Importance:

- Generalizable theory for multi-dimensional scanning using tessellating pattern
- Ideal platform to integrate with active-feedback tracking systems

Project 2: 3D-TrIm – Simultaneous Single-Virus Tracking and 3D Imaging

Aims:

- Develop new microscope capable of single virus tracking with simultaneous 3D imaging of surrounding environment.

Results:

- Developed 3D-TrIm microscope which generates volume spaces with co-registered single particle trajectories and 3D cellular imaging

Importance:

- First Observation of virus “skimming” behavior
- Discovered distinct diffusional modes on actin-rich protrusions
- Extension of virus tracking to live tissue systems demonstrated using multi-layer epithelial cells.

Texas Woman’s University, Denton, TX

Advisor: Prof. Richard D. Sheardy, Ph.D.

Studied structure and physical properties of i-motif DNA conformation using spectroscopic (UV-Vis, and circular dichroism) and calorimetric (DSC, ITC) instrumental methods to evaluate sequence context of conformation physical and chemical properties.

Publications

- High-speed 3D single-virus tracking reveals actin-aided viral trafficking of SARS-CoV-2 on the plasma membrane** 2026
Yuxin Lin, Xinrui Lu, Jack Exell, Haoting Lin, [Courtney Johnson](#), Kevin D. Welsher
bioRxiv **2026**
DOI: [10.48550/arXiv.2504.14157](#) (preprint)
- Phase diversity wavefront sensing for adaptive optics light-sheet fluorescence microscopy** 2026
[Courtney Johnson](#), Magdalena C. Schneider, Ryan Christensen, Xuesong Li, Min Guo, Patrick La Riviere, Hari Shroff
Proc. SPIE 13860, Adaptive Optics and Wavefront Control for Biological Systems XII **2026**
DOI: [10.48550/arXiv.2504.14157](#)
- DeepPD: Joint Phase and Object Estimation from Phase Diversity with Neural Calibration of a Deformable Mirror** 2025
Magdalena C. Schneider, [Courtney Johnson](#), Cedric Allier, Larissa Heinrich, Diane Adjavon, Joren Husic, Patrick La Rivière, Stephan Saalfeld, Hari Shroff
arXiv **2025**
DOI: [10.48550/arXiv.2504.14157](#) (preprint)
- Phase diversity-based wavefront sensing for fluorescence microscopy** 2024
[Courtney Johnson](#), Min Guo, Magdalena C. Schneider, Yijun Su, Satya Khuon, Nikolaj Reiser, Yicong Wu, Patrick La Riviere, and Hari Shroff
Optica **2024**
DOI: [10.1364/OPTICA.518559](#)
- Capturing the start point of the virus-cell interaction with high-speed 3D single-virus tracking** 2022
[Courtney Johnson](#), Jack Exell, Jonathan Aguilar, and Kevin Welsher
Nature Methods **2022**, 19, 1642–1652
DOI: [10.1038/s41592-022-01672-3](#)
- Continuous Focal Translation Enhances Rate of Volumetric Imaging** 2019
[Courtney Johnson](#), Jack Exell, Jonathon Kuo, and Kevin Welsher
Opt. Express **2019** 27, 36241–36258
DOI: [10.1364/OE.27.036241](#)
- Real-Time 3D Single Particle Tracking: Towards Active Feedback Single Molecule Spectroscopy in Live Cells** 2019
Shangguo Hou, [Courtney Johnson](#), and Kevin Welsher
Molecules **2019**, 24(15), 2826
DOI: [10.3390/molecules24152826](#)
- Point-Scan Volumetric Imaging Rate Increased by an Optimized Linear Sparse Sampling Pattern** 2019
[Courtney Johnson](#) and Kevin Welsher
Frontiers in Optics + Laser Science APS/DLS **2019** OSA Technical Digest (Optical Society of America), paper JTu4A.101.
DOI: [10.1364/FIO.2019.JTu4A.101](#)
- Loop Sequence Context Influences the Formation and Stability of the i-Motif for DNA Oligomers of Sequence (CCCXX)₄, where X = A and/or T, under Slightly Acidic Conditions** 2016
Mikeal McKim, Alexander Buxton, [Courtney Johnson](#), Amanda Metz, and Richard D. Sheardy
The Journal of Physical Chemistry B **2016** 120 (31), 7652–7661
DOI: [10.1021/acs.jpcb.6b04561](#)

Presentations

Invited Speaker – FIO+LS – Rochester, NY (forthcoming)	Sep. 2026
Invited Speaker – ASBMB 2026 – National Harbor, MD	Mar. 2026
Invited Speaker – SPIE BIOS Adaptive Optics – San Francisco, CA	Jan. 2026
Invited Speaker (International) – Optica Imaging Congress, (Adaptive Optics: Methods, Analysis and Applications) – Toulouse, France	Jul. 2024
HHMI January 2024 Science Meeting (Poster)	Jan. 2024
Invited Speaker/Panelist – Mathworks Bio Roundtable Discussion	Jan. 2024
Invited Speaker – SERMACS 2023 (Single-molecule Dynamics in Complex Chemical and Biological System Symposium) – Durham, NC	Oct. 2023
Selected Speaker – Janelia Symposium 2023	Jan. 2023
Janelia/EMBL Optical Interest Group	Jan. 2023
<i>High-Speed 3D Tracking and Imaging Microscopy Captures Early Events of the Virus-Cell Interaction</i>	
Invited Speaker/Panelist – TWU Celebration of Women in Science – Denton, TX	Oct. 2022
Invited Speaker – TWU Jane Nelson Institute for Women's Leadership – Denton, TX	May 2022
Invited Speaker – Southeast Biophysics Consortium Meeting – Denton, TX	May 2022
Selected Speaker – Fitzpatrick Institute of Photonics Seminar Series	Mar. 2022
SPIE Photonics West (Poster) – San Francisco, CA	Jan. 2022
Just Another (Chemistry) Webinar series (www.JAWSchem.com)	Nov. 2021
<i>"Multi-Scale 3D Visualization of the Cellular Landscape through Single-Virus Tracking"</i>	
Duke Chemistry Research Symposium (Talk)	Sep. 2021
1st Prize Poster – Fitzpatrick Institute of Photonics Annual Meeting	May 2021
<i>"Real-Time 3D Tracking and Imaging Microscopy"</i>	
Invited Speaker - 65 th Annual Meeting of the Biophysical Society (Nanoscale Approaches to Biology Subgroup Symposium)	Feb. 2021
<i>"Synchronous 3D Tracking and Imaging at Multiple Scales to Overcome Spatiotemporal Disparity"</i>	
65 th Annual Meeting of the Biophysical Society (Poster)	Feb. 2021
Fitzpatrick Institute of Photonics Annual Meeting (Poster)	Mar. 2020
Invited Speaker – 259 th ACS National Meeting (ACS Award for Encouraging Women into Careers in the Chemical Sciences: Symposium in honor of Katherine Franz)	Mar. 2020
Duke Chemistry Research Symposium (Poster)	Oct. 2019
ALIS Kickoff and Southeast Ultrafast Laser Conference	Oct. 2019
Frontiers in Optics + Laser Science (Lighting Talk + Poster) – Washington, DC	Sep. 2019
Duke Chemistry Research Symposium (Talk)	Oct. 2018
Fitzpatrick Institute of Photonics Annual Meeting (Poster)	Mar. 2018
Invited Speaker – STEM Education in Mississippi: Issues and Innovations – Jackson, MS	Oct. 2017
Invited Speaker/Panelist – TWU Celebration of Women in Science – Denton, TX	Oct. 2016
Duke Chemistry Research Symposium (Poster)	Oct. 2016
247 th American Chemical Society National Meeting (Poster) – Dallas, TX	Mar. 2014
Workshop Facilitator	Dec. 2014
<i>Designing a SENCER Course (UT San Antonio) – San Antonio, TX</i>	
Commendation of Excellence – National Society for Experiential Education 43 rd Annual Conference (Poster) – Baltimore, MD	Oct. 2014
TWU Quality Enhancement Program Experiential Scholar Program (Talk)	Apt. 2014

Grants and Scholarships

Joe Taylor Adams Fellowship (Duke Chemistry Department)	Fall 2021
Marcus Hobbs Fellowship (Duke Chemistry Department)	Spring 2020
TWU S-STEM QuaSARS Program - Full Scholarship (Tuition, Fees, Textbooks)	2013-2015
Robert A. Welch Foundation Grant for Undergraduate Research in Chemistry	SU 2013, 2014
Helen Ludeman Scholarship (TWU Chemistry Department)	AY 2012, 2014
TWU Quality Enhancement Plan Experiential Student Scholar Award to develop a laboratory exercise for a freshman chemistry course.	SU 2014
TWU Undergraduate Research Microgrant Program Award	Spring 2015

Awards and Honors

Leading Edge Fellow	2024
1 st Prize Student Speaker for Fitzpatrick Institute of Photonics Seminar Series	2022
1st Prize Poster at Fitzpatrick Institute of Photonics Annual Meeting 2021	2021
TWU Chancellor's Student Research Scholar Award for outstanding achievement in research.	2015
Commendation of Excellence for poster presentation "Practice Makes Perfect Proficient: Reflections on the Impact of Experiential Learning" (from the National Society for Experiential Education)	2014
TWU Chancellor's List for achieving 4.0 GPA with full-time student status	2010-2015
Nominated to membership in the Honor Society of Phi Kappa Phi	2014

Professional Development Activities

Duke Senior Leadership Insights Interviewer	Oct. 2021
Selected by the assistant dean to interview a senior executive for a professional development seminar featured on the main Duke Youtube page .	
Nationally competitive University of Michigan NextProf Science Workshop	May 2021
Workshop for promising future faculty candidates.	
Duke Preparing Future Faculty Program	AY 2020
Prepared and delivered guest lecture for Analytical Chemistry at NCCU	
"Unity in Diversity – Leading Through a Crisis"	August 2020
Published blog post, Duke Professional Development Blog	
Duke Emerging Leaders Institute Program	Spring 2020
Competitive application process selects only 20 total postdocs and grad students campus-wide to study leadership, culminating in a team project.	
"Letting Go, and Other Lessons from the Mat"	July 2019
Published blog post, Duke Professional Development Blog	

Teaching Experience

Duke University	2015-2019
<i>Graduate Teaching Assistant</i>	
CHEM 210 – Modern Applications of Chemistry Laboratory	
CHEM 202 – Organic Chemistry II Laboratory	
CHEM 110 – Core Concepts in Chemistry Discussion	
<i>Grader</i>	2015-2016
CHEM 201 – Organic Chemistry I Lecture	
<i>Tutor</i>	2016-2019

CHEM 101 – General Chemistry, Duke Athletics Tutor Program. Designed individualized program of instruction for 20+ student-athletes. Developed and taught 8-week summer preparation course for a small group of athletes taking CHEM101 in the fall.

Texas Woman's University

Undergraduate Teaching Assistant

2012- 2015

CHEM 1011 – Introduction to Chemistry Laboratory

CHEM 1111 – General Chemistry I Laboratory

CHEM 1121 – General Chemistry II Laboratory

Grader

2012, 2014

CHEM 1013 – Introduction to Chemistry Lecture

Service and Outreach Activities

Science Beyond the Bench

Spring 2023

Gave a talk at Science National Honor Society meeting at Broad Run High School

Regeneron Science Talent Search

Spring 2023

Gave a talk to finalists of the Regeneron Science Talent Search during their tour of Janelia.

Duke Women in Science and Engineering (WiSE)

Fall 2020

Invited to plan and organize women's self-defense programming

Duke Brazilian Jiu-Jitsu Club

2019-Present

Elected Safety Officer

Ensure safe behavior and assist with first aid for injuries of athletes

Duke Graduate Chemistry Council

2016-2018

Elected Media Coordinator

Promoted GCC activities and outreach events through social media and photographed events.

Kappa Epsilon Mu "KEM Club"

2014

Presented Chemistry Magic Shows at Events

Assisted in Preparation of Chemical Reagents for Magic Shows.

Professional Certifications

CompTIA A+ Certified IT Professional

2006

Certified to build, troubleshoot and repair Windows-based PCs.

Certificate verification ID: [N959CCKC6KQE5WRM](#)

Professional Society Memberships

American Chemical Society

Optical Society of America

American Physical Society

Biophysical Society

SPIE